

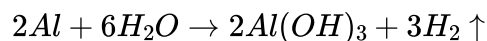
ES-MEGA-ALU ENERGY SYSTEM

TECHNICAL DOSSIER (ENGLISH)

Formulas, Derivations, Reaction Mechanics, Energy Paths

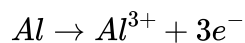
1. Chemical Fundamentals

1.1 Overall Reaction



2. Electrical Energy Generation

Electron release:



Charge per mol:

$$Q = 3F = 289455\text{ C}$$

Electrical energy:

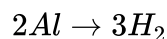
$$E = U \cdot Q$$

With 1.2–1.4 V:

$$E_{el} \approx 3.8\text{--}4.2\text{ kWh/kg}$$

3. Hydrogen Production

Reaction:



1 kg Al $\rightarrow$ 1.25 m³ H₂

Energy:

$$E_{H_2} \approx 1.8\text{--}2.2\text{ kWh/kg}$$

4. Thermal Energy

Reaction enthalpy:

$$\Delta H \approx -850 \text{ kJ/mol}$$

Thermal output:

$$E_{th} \approx 1\text{--}1.5 \text{ kWh/kg}$$

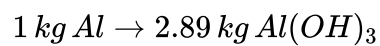
5. Total Energy per kg Aluminum

$$E_{total} = E_{el} + E_{H_2} + E_{th}$$

$$E_{total} \approx 6.6\text{--}7.7 \text{ kWh/kg}$$

6. Material Streams

Hydroxide:



7. Thermodynamics

Optimal temperature:

$$40\text{--}70^\circ\text{C}$$

8. Fluid Dynamics

Reynolds number:

$$Re = \frac{\rho v d}{\mu}$$

Laminar target:

$$Re < 2000$$

9. Electrolyte Flow

Flow rate:

$$\dot{V} = 8\text{--}15\text{ m}^3/h$$

Velocity:

$$v = \frac{\dot{V}}{A}$$

10. Power Output per Module

Example: 30,000 kg Al

$$E_{el} = 120,000\text{ kWh}$$

$$E_{H_2} = 60,000\text{ kWh}$$

$$E_{th} = 36,000\text{ kWh}$$

Total:

$$\approx 20\text{--}24\text{ MW}$$

11. Safety Formulas

Ideal gas law:

$$pV = nRT$$

Max overpressure:

$$p < 0.3\text{ bar}$$

End of Technical Dossier (English)